Jafar Abbaszadeh Chekan

PhD in Aerospace Engineering · University of Illinois Urbana-Champaign

1308 W Main St, CSL Building Rm 360, Urbana, IL 61801, USA

jafar2@illinois.edu | jafar-abbaszadeh

Research Interests

My research focuses on **decision-making under uncertainty**, integrating ideas from **online learning**, **optimization**, **control theory**, and **game theory**, with an emphasis on finite-time statistical and system-theoretic guarantees. My doctoral research develops regret-guaranteed learning algorithms for dynamical systems, including an anytime OFU-based algorithm for LQ control with unknown dynamics, and investigates the interplay between online learning and switched/hybrid systems control. My current interests extend toward multi-agent learning and sequential decision-making problems with mixed discrete and continuous decisions. On the application side, I am particularly interested in autonomous and multi-agent systems, including robotics, as well as high-stakes sequential decision-making settings such as social systems and healthcare.

Education

University of Illinois Urbana-Champaign — *Urbana, IL, USA*

PhD in Aerospace Engineering — *Jan 2019–Sep 2026*

- Thesis: Safety-Aware Learning-Based Control
- Advisors: Prof. Cedric Langbort and Assoc. Prof. S. Rasoul Etesami (co-advisor)
- Doctoral committee: Cedric Langbort (UIUC, Chair), S. Rasoul Etesami (UIUC, Co-Chair), Jeff Shamma (UIUC), Na Li (Harvard), Karan Singh (CMU), Nan Jiang (UIUC)
- GPA: 3.80/4.00

Virginia Polytechnic Institute and State University — *Blacksburg, VA, USA*

MSc in Engineering Mechanics — *Aug 2015–Feb 2018*

- Thesis: Data-Driven Optimal Control and Power Management of Energy Systems
- GPA: 3.92/4.00

Sharif University of Technology — *Tehran, Iran*

MSc in Mechanical Engineering — *Sep 2008–Jun 2011*

- Thesis: Control of Chaotic Systems; Stability Analysis of Dynamical Systems
- GPA: 15.42/20

Bahonar University of Kerman — *Kerman, Iran*

BSc in Mechanical Engineering — *Jan 2004–Sep 2008*

- GPA: 15.22/20

Publications

Journal Articles

J. Abbaszadeh Chekan and C. Langbort, "Learn and Control while Switching: With Guaranteed Stability and Sublinear Regret," *IEEE Transactions on Automatic Control*, vol. 69, no. 12, pp. 8433–8448, 2024. doi: 10.1109/TAC.2024.3440348.

J. Abbaszadeh Chekan and C. Langbort, "Regret Bounds for LQ Adaptive Control Under Database Attacks," *Automatica*, vol. 151, p. 110876, 2023. doi: 10.1016/j.automatica.2023.110876.

Conference Articles

J. Abbaszadeh Chekan and C. Langbort, "Safety-Aware Learning-Based Control of Systems with Uncertainty Dependent Constraints," in *Proc. American Control Conference (ACC)*, 2024.

J. Abbaszadeh Chekan and C. Langbort, "Joint Stabilization and Regret Minimization through Switching in Over-Actuated Systems," in *Proc. IFAC Symposium on Robust Control Design (ROCOND)*, 2024.

Preprints and Papers Under Review

J. Abbaszadeh Chekan and C. Langbort, "Any-Time Regret-Guaranteed Algorithm for Control of Linear Quadratic Systems," arXiv:2406.07746, 2025. *Under review, Journal of Machine Learning Research.*

J. Abbaszadeh Chekan, S. R. Etesami, and C. Langbort, "Regret-Guaranteed Safe Switching with Minimum Cost: LQR Setting with Unknown Dynamics," arXiv:2606.22223, 2026. *Under review, SIAM Journal on Control and Optimization (SICON).*

In Preparation

J. Abbaszadeh Chekan, S. R. Etesami, and C. Langbort, "Safe Learning-Based Control: Higher-Order Control Barrier Functions Certification," *in preparation*.

Earlier Work in Applied Control and Energy Systems

J. Abbaszadeh Chekan, M. A. Nojoumian, K. Merat, and H. Salarieh, "Chaos Control in Lateral Oscillations of Spinning Disk via Linear Optimal Control of Discrete Systems," *Journal of Vibration and Control*, vol. 23, no. 1, pp. 103–110, 2017.

K. Merat, **J. Abbaszadeh Chekan**, H. Salarieh, and A. Alasty, "Control of Discrete Time Chaotic Systems via Combination of Linear and Nonlinear Dynamic Programming," *ASME Journal of Computational and Nonlinear Dynamics*, vol. 10, no. 1, p. 011008, 2015. doi: 10.1115/1.4027716.

K. Merat, **J. Abbaszadeh Chekan**, H. Salarieh, and A. Alasty, "Linear Optimal Control of Continuous Time Chaotic Systems," *ISA Transactions*, vol. 53, no. 4, pp. 1209–1215, 2014. doi: 10.1016/j.isatra.2014.01.003.

J. Abbaszadeh Chekan, S. Bashash, and S. Taheri, "A Data-Driven Control Strategy for Trip Length-Conscious Power Management of Plug-In Hybrid Electric Vehicles," in *Proc. IEEE Conference on Control Technology and Applications (CCTA)*, pp. 794–799, 2018. doi: 10.1109/CCTA.2018.8511466.

J. Abbaszadeh Chekan and S. Bashash, "Dynamic Programming-Based Approximate Real-Time Control Policies for Plug-In Hybrid Electric Vehicles," in *Proc. IEEE Conference on Control Technology and Applications (CCTA)*, pp. 205–210, 2017. doi: 10.1109/CCTA.2017.8062464.

J. Abbaszadeh Chekan and S. Bashash, "IoT-Oriented Control of Aggregate Thermostatically-Controlled Loads," in *Proc. American Control Conference (ACC)*, pp. 4729–4734, 2017. doi: 10.23919/ACC.2017.7963686.

K. Merat, **J. Abbaszadeh Chekan**, H. Salarieh, and A. Alasty, "Online Hybrid Model Predictive Controller Design for Cruise Control of Automobiles," in *Proc. ASME Dynamic Systems and Control Conference (DSCC)*, pp. 5274–5283, 2017. doi: 10.1115/DSCC2017-5274.

J. Abbaszadeh Chekan, K. Merat, and H. Zohoor, "Stability Analysis of Rotating Stepped Shaft via Lyapunov Criterion," in *Proc. ASME International Mechanical Engineering Congress and Exposition (IMECE)*, pp. 701–706, 2012. doi: 10.1115/IMECE2012-89565.

Presentations

Invited Talks

"Fully Adaptive Regret-Guaranteed Algorithm for Control of Linear Quadratic Systems."
University of Pennsylvania, Philadelphia, PA, USA *November 2024*

"Fully Adaptive Regret-Guaranteed Algorithm for Control of Linear Quadratic Systems."
Massachusetts Institute of Technology, Cambridge, MA, USA *November 2024*

"Fully Adaptive Regret-Guaranteed Algorithm for Control of Linear Quadratic Systems."
Harvard University, Cambridge, MA, USA *November 2024*

"Fully Adaptive Regret-Guaranteed Algorithm for Control of Linear Quadratic Systems."
NetCon Seminar, KTH Royal Institute of Technology, Stockholm, Sweden *November 2024*

"How to Tune the Exploration–Exploitation Trade-off to Further Optimize Regret Bounds."
ETH Zurich, Zurich, Switzerland *October 2024*

"Fully Adaptive Regret-Guaranteed Algorithm for Control of Linear Quadratic Systems."
Informal Systems Seminar (ISS), McGill University, Montreal, Canada *September 2024*

"Regret Bounds for LQ Adaptive Control Under Database Attacks."
NetCon Seminar, KTH Royal Institute of Technology, Stockholm, Sweden *October 2022*

Contributed Presentations

"Fully Adaptive Regret-Guaranteed Convex SDP-Based Algorithm for Control of LQ Systems" (poster).
Graduation Day, Information Theory and Applications (ITA) Workshop, San Diego, CA, USA *February 2024*

"Safe Control of Partially Unknown Systems with Uncertainty Dependent Constraints" (oral).
Conference on Information Sciences and Systems (CISS), Johns Hopkins University, Baltimore, MD, USA *March 2023*

Teaching Experience

University of Illinois Urbana-Champaign

Graduate Teaching Assistant

- AE 498 – Review of Basic Mathematics: Topics in Linear Algebra and Real Analysis *Fall 2025*
- AE 461 – Structures and Control Lab *Spring 2024, Spring 2025*
- AE 504 – Optimal Aerospace Systems *Fall 2024*

Awards and Scholarships

2024 **University Block Grant Fellowship**, Grainger College of Engineering, UIUC
2023 **Mavis Future Faculty Fellowship**, Grainger College of Engineering, UIUC
2023 **Kenneth Lee Herrick Memorial Award**, Department of Aerospace Engineering, UIUC
2020 **Outstanding Graduate Student Award**, Department of Aerospace Engineering, UIUC

Professional Experience

Tower Foundation, San Jose State University *San Jose, CA, USA*

Visiting Researcher *Sep 2016–Aug 2017*

- Supervisor: Prof. Saeid Bashash
- Research: control of power and energy systems

Professional Service

Reviewer

Journals	IEEE Transactions on Automatic Control
	International Journal of Electrical Power and Energy Systems
Conferences	IFAC Symposium on Robust Control Design (ROCOND)
	American Control Conference (ACC)

Relevant Graduate Level Coursework

Decision and Control: Convex Methods in Control (ECE), Robust Control (AE), Linear Control Theory (ECE, AOE), Adaptive Control Systems (ECE, ME), Control of Stochastic Systems (ECE), Control of Complex Systems (SE), Stochastic Control for Motion and Mission Planning (AE)

Optimization: Optimization (ISE), Variational Methods (ESM), Optimization by Vector Space Methods (ECE), Topics in Game Theory and Fair Division (ISE)

Learning Theory & AI: Statistical Learning Theory (ECE), Statistical Reinforcement Learning (CS), Online Learning and Decision-Making Theory (ISE), Interplay Between Control and Machine Learning (ECE)

Mathematical and Systems Foundations: Applied Stochastic Processes (Math), Applied Tensor Analysis (ESM), Introduction to Continuum Mechanics (ESM), Advanced Engineering Mathematics (ME)

Skills

Programming Python (NumPy, SciPy, CVXPY), MATLAB, LaTeX

References

Cedric Langbort *Professor of Aerospace Engineering*,
University of Illinois Urbana-Champaign
+1 (217) 265-6236
langbort@illinois.edu

S. Rasoul Etesami *Associate Professor, Industrial & Enterprise Systems Engineering*,
University of Illinois Urbana-Champaign
+1 (217) 300-4793
etesami1@illinois.edu

Jeff Shamma *Professor and Department Head, Industrial & Enterprise Systems Engineering*,
University of Illinois Urbana-Champaign
+1 (217) 265-6210
jshamma@illinois.edu

Karan Singh *Assistant Professor of Operations Research*,
Carnegie Mellon University
karansin@andrew.cmu.edu

Nan Jiang *Associate Professor, Siebel School of Computing and Data Science*,
University of Illinois Urbana-Champaign
+1 (217) 300-8275
nanjiang@illinois.edu